Characterization of the PRs

omitted

2024-04-05

Data overview

```
glimpse(prs_characterization)
```

```
## Rows: 1,310
## Columns: 10
## $ pr_number          <int> 1302, 1304, 1313, 1322, 1329, 1366, 1367, 1381, 14…
## $ repo               <chr> "accumulo", "accumulo", "accumulo", "accumulo", "a…
## $ created_at         <dttm> 2019-07-31 19:34:39, 2019-08-01 18:52:50, 2019-08…
## $ merged_at          <dttm> 2019-08-02 15:59:29, 2019-08-01 20:56:42, 2019-08…
## $ additions          <int> 235, 89, 555, 112, 404, 363, 97, 961, 5, 2, 7, 301…
## $ deletions          <int> 348, 91, 521, 109, 255, 19, 20, 10, 33, 0, 27, 258…
## $ changed_files_count <int> 15, 5, 20, 5, 16, 7, 7, 11, 20, 1, 1, 2132, 38, 28…
## $ commits_count      <int> 4, 2, 3, 1, 4, 4, 5, 3, 1, 1, 2, 4, 1, 2, 2, 1, 7,…
## $ code_churn         <int> 583, 180, 1076, 221, 659, 382, 117, 971, 38, 2, 34…
## $ duration           <dbl> 1.850578704, 0.086018519, 6.993090278, 1.359756944…
```

PRs count

```
prs_characterization %>% count()
```

```
##      n
## 1 1310
```

PRs count by repo

```
prs_characterization %>%
    group_by(repo) %>%
    count()
```

```
## # A tibble: 12 × 2
## # Groups:   repo [12]
##    repo                      n
##    <chr>                 <int>
##  1 accumulo                573
##  2 cayenne                  46
##  3 commons-collections      25
##  4 commons-io               64
##  5 commons-lang             69
##  6 helix                   217
##  7 httpcomponents-client    72
##  8 maven-surefire           22
##  9 opennlp                  66
## 10 struts                  100
## 11 wicket                   33
## 12 zookeeper                23
```

Commits count

```
sum(prs_characterization$commits_count)
```

```
## [1] 3121
```

Commits count by repo

```
prs_characterization %>%
    group_by(repo) %>%
    summarise(sum(commits_count))
```

```
## # A tibble: 12 × 2
##    repo                  `sum(commits_count)`
##    <chr>                                <int>
##  1 accumulo                              1277
##  2 cayenne                                139
##  3 commons-collections                     51
##  4 commons-io                             171
##  5 commons-lang                           149
##  6 helix                                  678
##  7 httpcomponents-client                  135
##  8 maven-surefire                          34
##  9 opennlp                                 77
## 10 struts                                 274
## 11 wicket                                  78
## 12 zookeeper                               58
```

General data statistics by repo

```
prs_characterization %>%
    group_by(repo) %>%
    summarise(
        min_date = min(merged_at),
        max_date = max(merged_at),
        age_in_days = difftime(min(merged_at), max(merged_at), units = "days")
    ) %>%
    mutate(
    age_in_days = abs(as.numeric(age_in_days)),
    age_in_years = abs(as.numeric(age_in_days))/365.25
    )
```

```
## # A tibble: 12 × 5
##    repo         min_date            max_date            age_in_days age_in_years
##    <chr>        <dttm>              <dttm>                    <dbl>        <dbl>
##  1 accumulo     2019-08-01 20:56:42 2024-02-29 23:40:37       1673.         4.58
##  2 cayenne      2021-12-07 14:22:09 2024-02-06 07:24:10        791.         2.16
##  3 commons-col… 2019-02-17 08:09:56 2024-01-20 15:02:23       1798.         4.92
##  4 commons-io   2016-12-02 02:08:46 2024-02-09 22:53:32       2626.         7.19
##  5 commons-lang 2016-11-22 21:25:59 2024-01-30 18:34:19       2625.         7.19
##  6 helix        2020-10-06 21:58:38 2024-02-27 18:14:32       1239.         3.39
##  7 httpcompone… 2018-12-23 22:09:39 2024-02-29 16:12:35       1894.         5.18
##  8 maven-suref… 2021-11-18 22:43:22 2023-08-23 16:41:09        643.         1.76
##  9 opennlp      2017-04-10 09:32:10 2024-01-13 13:53:26       2469.         6.76
## 10 struts       2018-12-06 09:08:03 2024-02-13 15:18:04       1895.         5.19
## 11 wicket       2019-10-09 13:06:17 2024-02-26 11:33:05       1601.         4.38
## 12 zookeeper    2022-01-26 12:23:10 2024-02-12 18:40:34        747.         2.05
```

```
prs_characterization %>%
    group_by(repo) %>%
    summarise(
        mean_commits = mean(commits_count),
        median_commits = median(commits_count),
        mean_changed_files = mean(changed_files_count),
        median_changed_files = median(changed_files_count)
    )
```

```
## # A tibble: 12 × 5
##    repo      mean_commits median_commits mean_changed_files median_changed_files
##    <chr>            <dbl>          <dbl>              <dbl>                <dbl>
##  1 accumulo          2.23              2              14.9                   4
##  2 cayenne           3.02              2               9.13                  4.5
##  3 commons-…         2.04              2              15.7                   4
##  4 commons-…         2.67              2               3.45                  2
##  5 commons-…         2.16              1               4.14                  2
##  6 helix             3.12              2               6.01                  3
##  7 httpcomp…         1.88              1              10.8                   3
##  8 maven-su…         1.55              1              24.7                   3
##  9 opennlp           1.17              1              12.3                   4
## 10 struts            2.74              2              16.1                   4
## 11 wicket            2.36              1               4.24                  2
## 12 zookeeper         2.52              2               7.26                  3
```

```
prs_characterization %>%
    group_by(repo) %>%
    summarise(
        mean_code_churn = mean(code_churn),
        median_code_churn = median(code_churn),
        mean_duration = mean(duration)
    )
```

```
## # A tibble: 12 × 4
##    repo                  mean_code_churn median_code_churn mean_duration
##    <chr>                           <dbl>             <dbl>         <dbl>
##  1 accumulo                        510.               97            3.01
##  2 cayenne                         435.               92            6.61
##  3 commons-collections             164.               56           48.8
##  4 commons-io                       83.3              30.5         14.2
##  5 commons-lang                    156.               34           16.1
##  6 helix                           234.               77            7.95
##  7 httpcomponents-client           394.               59            1.86
##  8 maven-surefire                  498.               31           18.9
##  9 opennlp                         362.               95            2.61
## 10 struts                         1394.              174.           7.50
## 11 wicket                          111.               62            5.07
## 12 zookeeper                       155.               86           58.1
```

NCLOC and classes count by repo

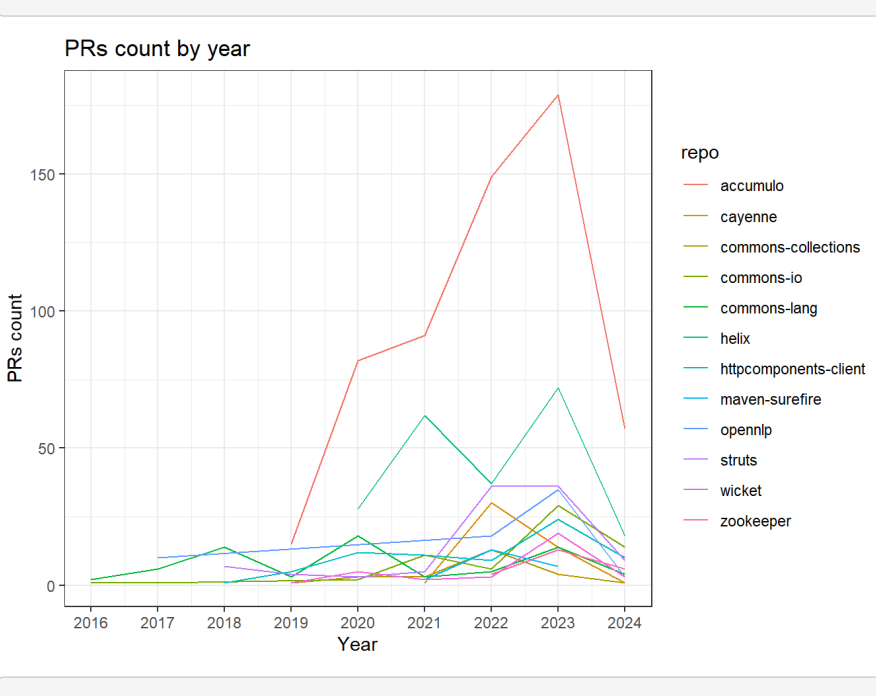
- All values are related to the last commit of the last PR that was successfully executed by SonarQube.

```
data.frame(repo = c("accumulo", "cayenne", "commons-collections","commons-io", "commons-lang", "helix", "httpcomponents-clie
nt", "maven-surefire", "opennlp", "struts", "wicket", "zookeeper"), NCLOC = c(440441, 318428, 67690, 30501, 95929, 189487, 7
6964, 110481, 155900, 234331, 251505, 131712), classes = c(5164, 4716, 839, 288, 918, 2106, 879, 3036, 2478, 3419, 5254, 154
2))
```

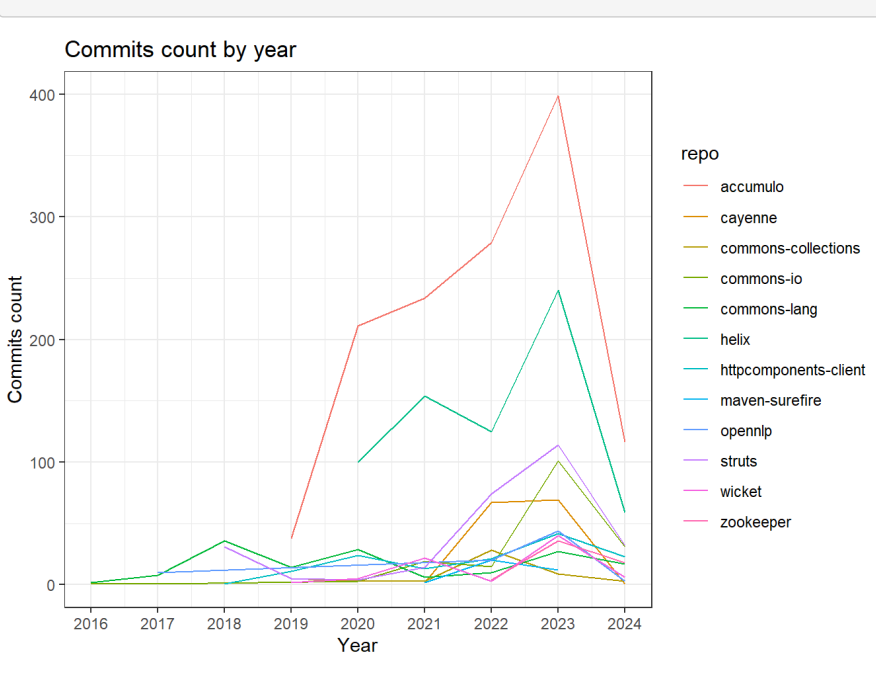
```
##                     repo  NCLOC classes
## 1               accumulo 440441    5164
## 2                cayenne 318428    4716
## 3    commons-collections  67690     839
## 4             commons-io  30501     288
## 5           commons-lang  95929     918
## 6                  helix 189487    2106
## 7  httpcomponents-client  76964     879
## 8         maven-surefire 110481    3036
## 9                opennlp 155900    2478
## 10                struts 234331    3419
## 11                wicket 251505    5254
## 12             zookeeper 131712    1542
```

PRs and commits count through the years

```
prs_characterization %>%
    mutate(year = lubridate::year(merged_at)) %>%
    group_by(year, repo) %>%
    summarise(PRs_quantity = n(), .groups = 'drop') %>%
    ggplot(aes(x = year, y = PRs_quantity, color=repo)) +
    geom_line() +
    labs(x = "Year", y = "PRs count") +
    ggtitle("PRs count by year") +
    scale_x_continuous(breaks = unique(lubridate::year(prs_characterization$merged_at)))
```



```
prs_characterization %>%
    mutate(year = lubridate::year(merged_at)) %>%
    group_by(year, repo) %>%
    summarise(commits_quantity = sum(commits_count), .groups = 'drop') %>%
    ggplot(aes(x = year, y = commits_quantity, color=repo)) +
    geom_line() +
    labs(x = "Year", y = "Commits count") +
    ggtitle("Commits count by year") +
    scale_x_continuous(breaks = unique(lubridate::year(prs_characterization$merged_at)))
```



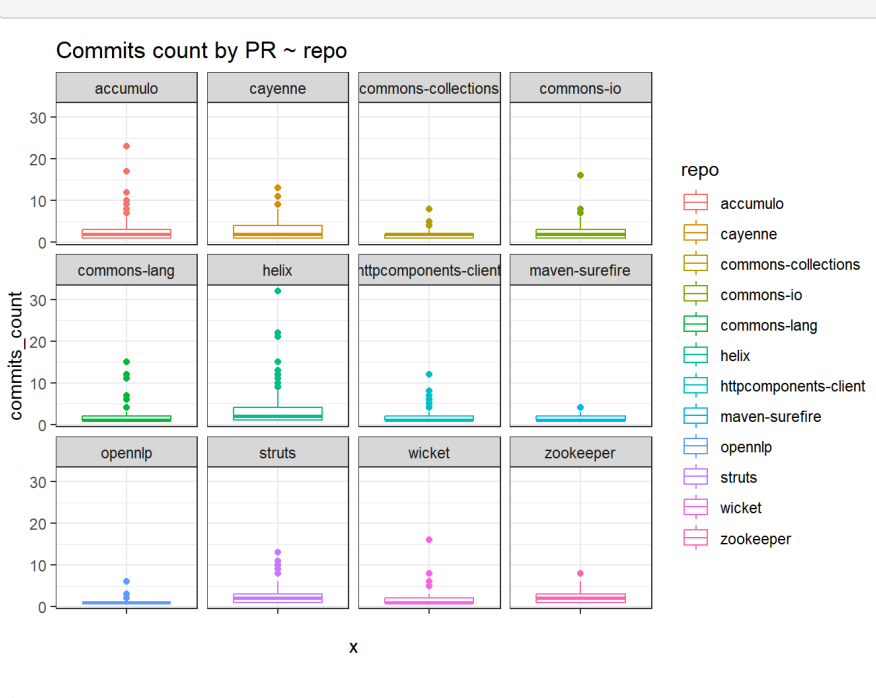
Distribution of features

Commits count by PR

```
tapply(prs_characterization$commits_count, prs_characterization$repo, summary)
```

```
## $accumulo
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   2.000   2.229   3.000  23.000 
## 
## $cayenne
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   2.000   3.022   4.000  13.000 
## 
## $`commons-collections`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    1.00    2.00    2.04    2.00    8.00 
## 
## $`commons-io`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   2.000   2.672   3.000  16.000 
## 
## $`commons-lang`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   1.000   2.159   2.000  15.000 
## 
## $helix
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   2.000   3.124   4.000  32.000 
## 
## $`httpcomponents-client`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   1.000   1.875   2.000  12.000 
## 
## $`maven-surefire`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   1.000   1.545   2.000   4.000 
## 
## $opennlp
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   1.000   1.167   1.000   6.000 
## 
## $struts
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    1.00    2.00    2.74    3.00   13.00 
## 
## $wicket
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   1.000   2.364   2.000  16.000 
## 
## $zookeeper
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   2.000   2.522   3.000   8.000
```

```
prs_characterization %>%
    ggplot(aes(x = "", y = commits_count, color=repo)) +
      geom_boxplot() +
      facet_wrap(~repo) +
      labs(title = "Commits count by PR ~ repo")
```

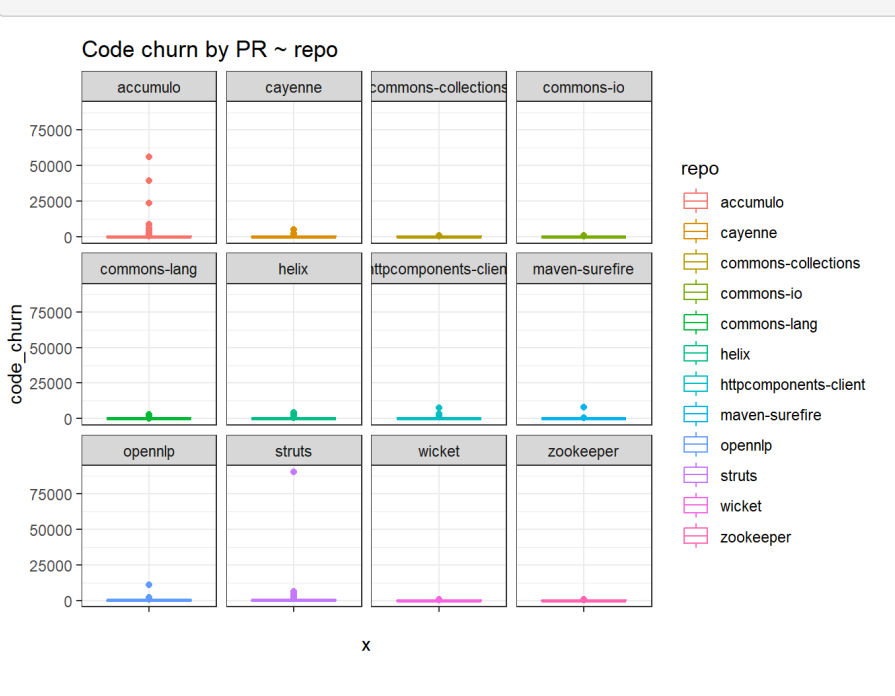


Code churn by PR

```
tapply(prs_characterization$code_churn, prs_characterization$repo, summary)
```

```
## $accumulo
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     1.0    30.0    97.0   509.9   245.0 55972.0 
## 
## $cayenne
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     2.0    31.0    92.0   435.1   470.5  4793.0 
## 
## $`commons-collections`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##       2      11      56     164     168    1035 
## 
## $`commons-io`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00   15.50   30.50   83.31   65.50  631.00 
## 
## $`commons-lang`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     2.0    11.0    34.0   155.9   108.0  2696.0 
## 
## $helix
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##       1      24      77     234     209    4343 
## 
## $`httpcomponents-client`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     2.0    12.0    59.0   393.8   272.2  7633.0 
## 
## $`maven-surefire`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    2.00   13.75   31.00  497.91  171.75 8229.00 
## 
## $opennlp
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max. 
##     2.00    26.25    95.00   361.50   220.00 11178.00 
## 
## $struts
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     2.0    58.0   174.5  1394.3   479.0 90610.0 
## 
## $wicket
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     3.0    18.0    62.0   110.6   133.0   849.0 
## 
## $zookeeper
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     3.0    28.5    86.0   155.4   133.5   763.0
```

```
prs_characterization %>%
    ggplot(aes(x = "", y = code_churn, color=repo)) +
      geom_boxplot() +
      facet_wrap(~repo) +
      labs(title = "Code churn by PR ~ repo")
```

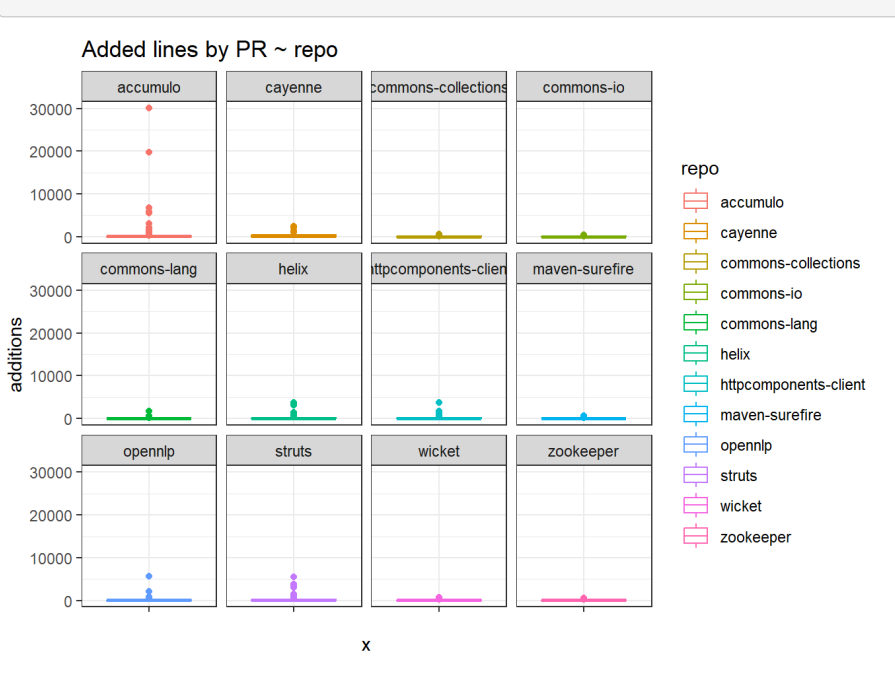


Added lines

```
tapply(prs_characterization$additions, prs_characterization$repo, summary)
```

```
## $accumulo
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0    14.0    50.0   256.4   142.0 30103.0 
## 
## $cayenne
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0    25.5    73.0   317.5   365.0  2447.0 
## 
## $`commons-collections`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    7.00   30.00   65.88   67.00  536.00 
## 
## $`commons-io`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    6.00   20.50   46.47   50.50  399.00 
## 
## $`commons-lang`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     1.0     6.0    20.0    92.7    65.0  1717.0 
## 
## $helix
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0    16.0    52.0   179.2   157.0  3718.0 
## 
## $`httpcomponents-client`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    5.75   41.50  218.94  153.75 3783.00 
## 
## $`maven-surefire`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0     4.0    13.0    79.5    66.5   579.0 
## 
## $opennlp
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     1.0    15.0    61.5   219.9   142.8  5676.0 
## 
## $struts
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00   25.75  113.50  340.01  239.25 5565.00 
## 
## $wicket
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    9.00   42.00   77.76   74.00  650.00 
## 
## $zookeeper
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     2.0    22.0    59.0   111.3   101.0   622.0
```

```
prs_characterization %>%
    ggplot(aes(x = "", y = additions, color=repo)) +
      geom_boxplot() +
      facet_wrap(~repo) +
      labs(title = "Added lines by PR ~ repo")
```

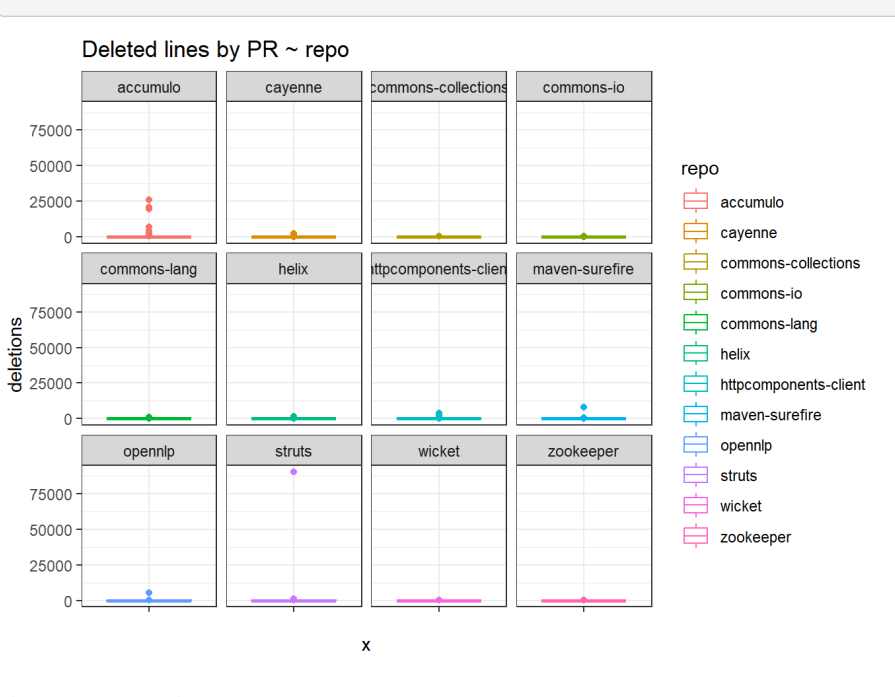


Deleted lines

```
tapply(prs_characterization$deletions, prs_characterization$repo, summary)
```

```
## $accumulo
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0     5.0    25.0   253.5    98.0 25869.0 
## 
## $cayenne
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0     2.0    16.5   117.6    49.5  2346.0 
## 
## $`commons-collections`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    1.00   15.00   98.16   84.00  557.00 
## 
## $`commons-io`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    1.00    7.00   36.84   21.25  423.00 
## 
## $`commons-lang`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    3.00    9.00   63.22   36.00  979.00 
## 
## $helix
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    3.00   13.00   54.73   37.00 1506.00 
## 
## $`httpcomponents-client`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    2.00    9.00  174.83   46.75 3850.00 
## 
## $`maven-surefire`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    5.50   12.50  418.41   26.75 8228.00 
## 
## $opennlp
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0     2.0    22.5   141.6    88.0  5502.0 
## 
## $struts
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0    12.0    52.0  1054.3   158.2 90609.0 
## 
## $wicket
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    4.00   11.00   32.82   27.00  199.00 
## 
## $zookeeper
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    2.00   18.00   44.13   56.50  307.00
```

```
prs_characterization %>%
    ggplot(aes(x = "", y = deletions, color=repo)) +
      geom_boxplot() +
      facet_wrap(~repo) +
      labs(title = "Deleted lines by PR ~ repo")
```

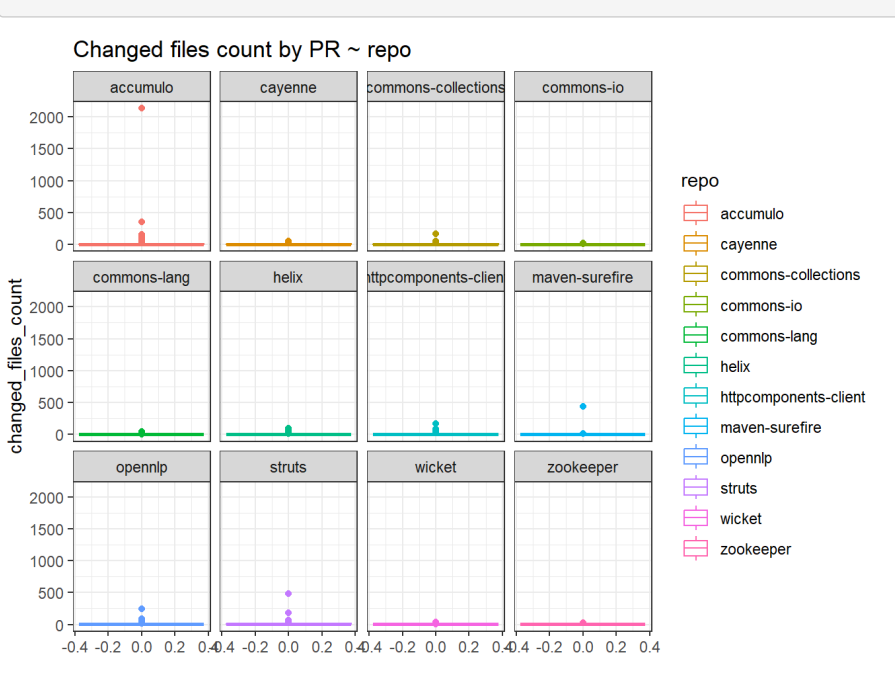


Changed files count

```
tapply(prs_characterization$changed_files_count, prs_characterization$repo, summary)
```

```
## $accumulo
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    2.00    4.00   14.86   10.00 2132.00 
## 
## $cayenne
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    2.00    4.50    9.13   11.00   57.00 
## 
## $`commons-collections`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    1.00    4.00   15.72   12.00  165.00 
## 
## $`commons-io`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   2.000   3.453   3.000  22.000 
## 
## $`commons-lang`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   2.000   4.145   3.000  42.000 
## 
## $helix
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   3.000   6.009   6.000  97.000 
## 
## $`httpcomponents-client`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    2.00    3.00   10.82    8.50  174.00 
## 
## $`maven-surefire`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    2.00    3.00   24.68    7.50  434.00 
## 
## $opennlp
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    2.00    4.00   12.26    8.75  242.00 
## 
## $struts
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    2.00    4.00   16.06   12.00  484.00 
## 
## $wicket
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   2.000   4.242   3.000  32.000 
## 
## $zookeeper
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   2.000   3.000   7.261  10.000  29.000
```

```
prs_characterization %>%
    ggplot(aes(y = changed_files_count, color=repo)) +
      geom_boxplot() +
      facet_wrap(~repo) +
      labs(title = "Changed files count by PR ~ repo")
```

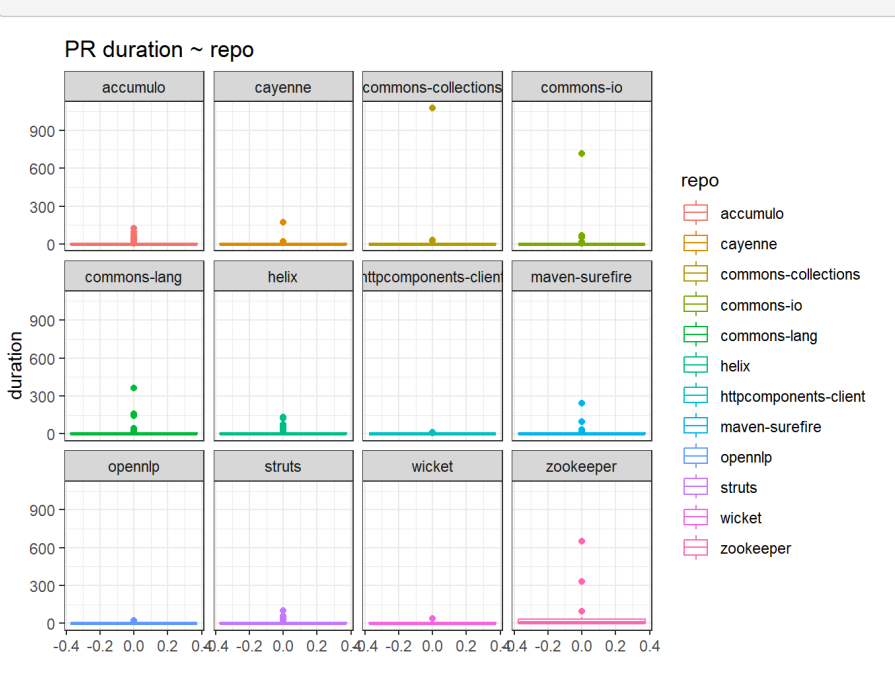


PR duration

```
tapply(prs_characterization$duration, prs_characterization$repo, summary)
```

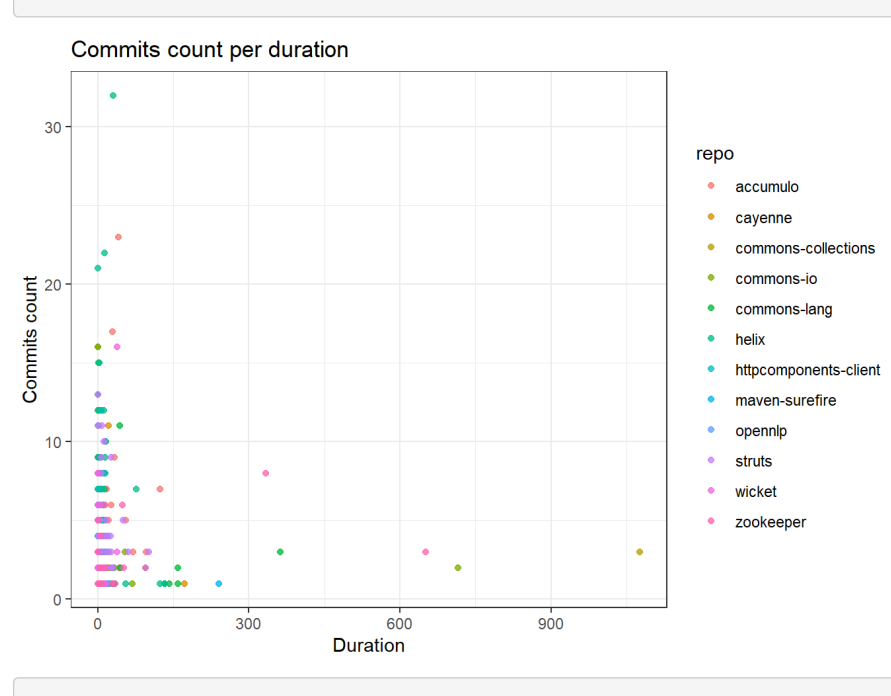
```
## $accumulo
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
## 7.292e-03 1.324e-01 7.998e-01 3.012e+00 2.814e+00 1.232e+02 
## 
## $cayenne
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
## 8.924e-03 2.181e-01 9.842e-01 6.611e+00 4.111e+00 1.717e+02 
## 
## $`commons-collections`
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
## 3.460e-03 1.751e-01 1.419e+00 4.879e+01 7.249e+00 1.076e+03 
## 
## $`commons-io`
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
##   0.02696   0.14277   0.38076  14.18991   1.12480 715.08228 
## 
## $`commons-lang`
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
##   0.01137   0.16159   0.74821  16.12686   6.35567 362.21278 
## 
## $helix
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
## 1.740e-04 8.057e-01 2.700e+00 7.952e+00 6.753e+00 1.328e+02 
## 
## $`httpcomponents-client`
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
##  0.002651  0.090764  0.761528  1.863347  2.948854 12.367778 
## 
## $`maven-surefire`
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max. 
##   0.1133   0.5496   1.3150  18.9145   4.2399 240.9541 
## 
## $opennlp
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max. 
##  0.00419  0.46633  1.13698  2.61029  2.67322 20.87878 
## 
## $struts
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
##   0.01865   1.02926   2.88605   7.50470   7.64988 101.57461 
## 
## $wicket
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max. 
##  0.03612  0.45771  1.37601  5.06871  5.89888 39.14480 
## 
## $zookeeper
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
##   0.04421   1.01348   8.83714  58.07788  31.23014 650.60612
```

```
prs_characterization %>%
    ggplot(aes(y = duration, color=repo)) +
      geom_boxplot() +
      facet_wrap(~repo) +
      labs(title = "PR duration ~ repo")
```

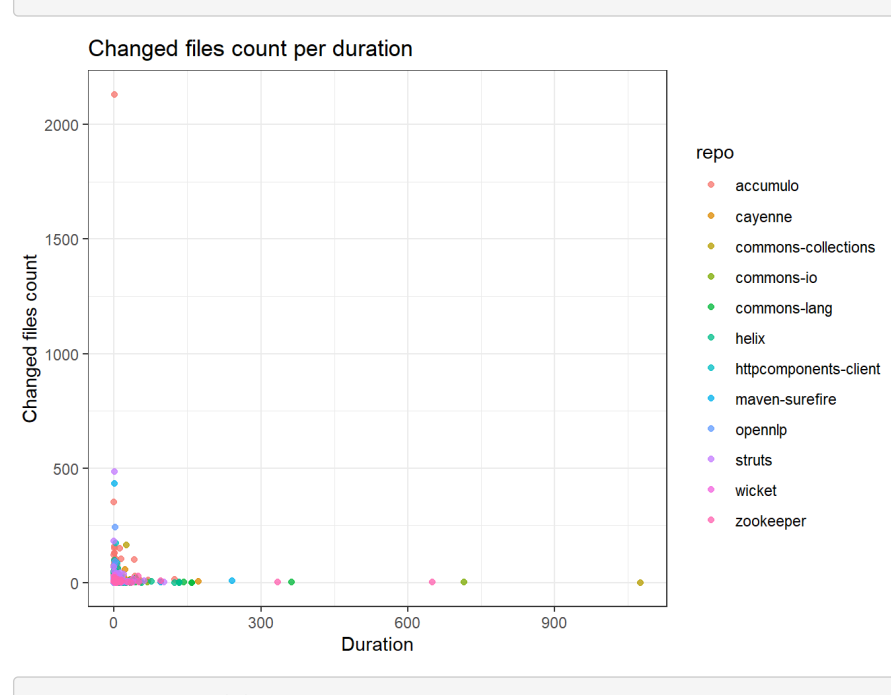


Scatters

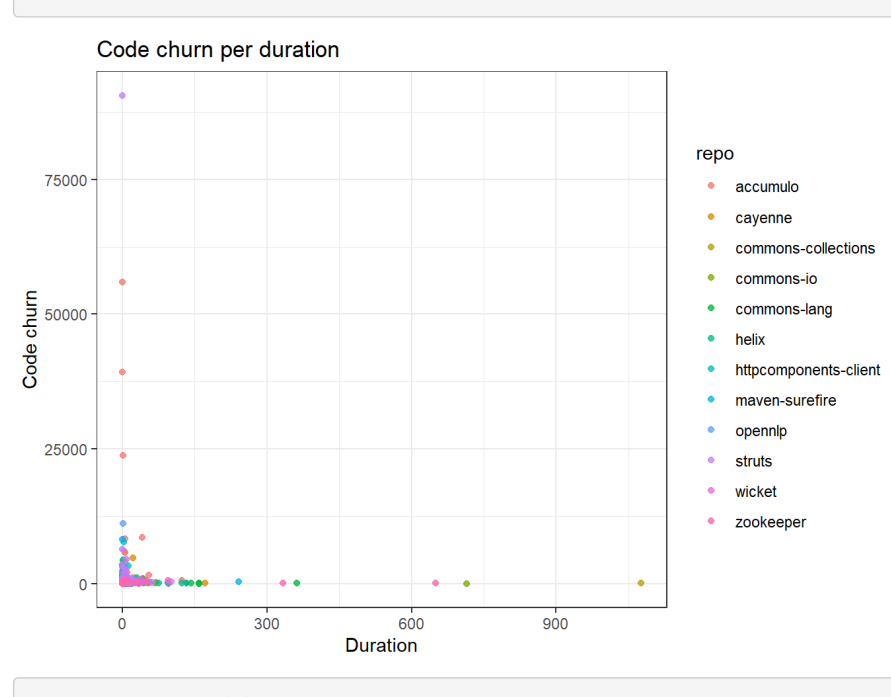
```
prs_characterization %>%
    ggplot(aes(x = duration, y = commits_count, color=repo)) +
    geom_point(alpha=0.75) +
    labs(x = "Duration", y = "Commits count") +
    ggtitle("Commits count per duration")
```



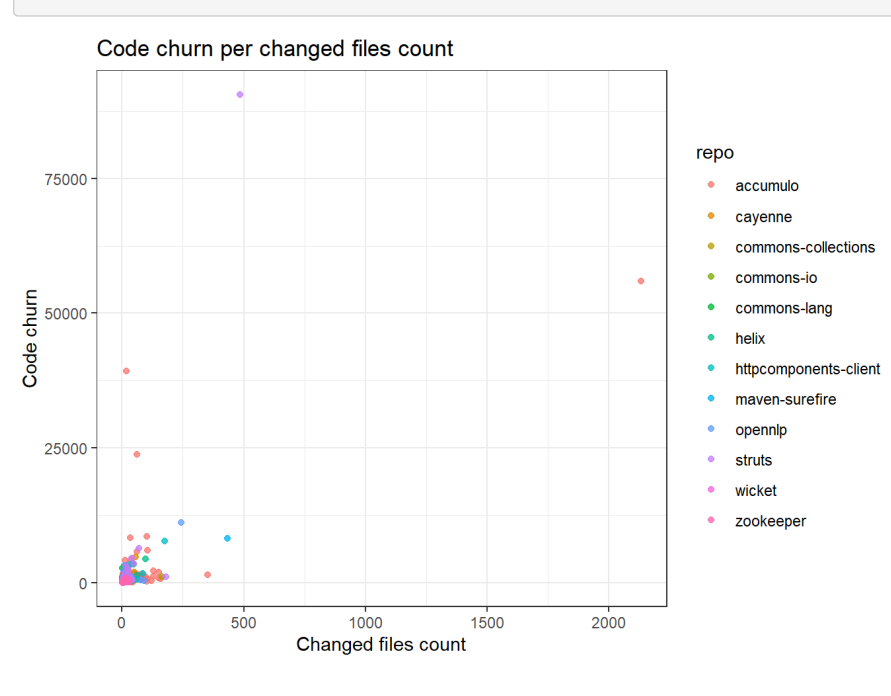
```
prs_characterization %>%
    ggplot(aes(x = duration, y = changed_files_count, color=repo)) +
    geom_point(alpha=0.75) +
    labs(x = "Duration", y = "Changed files count") +
    ggtitle("Changed files count per duration")
```



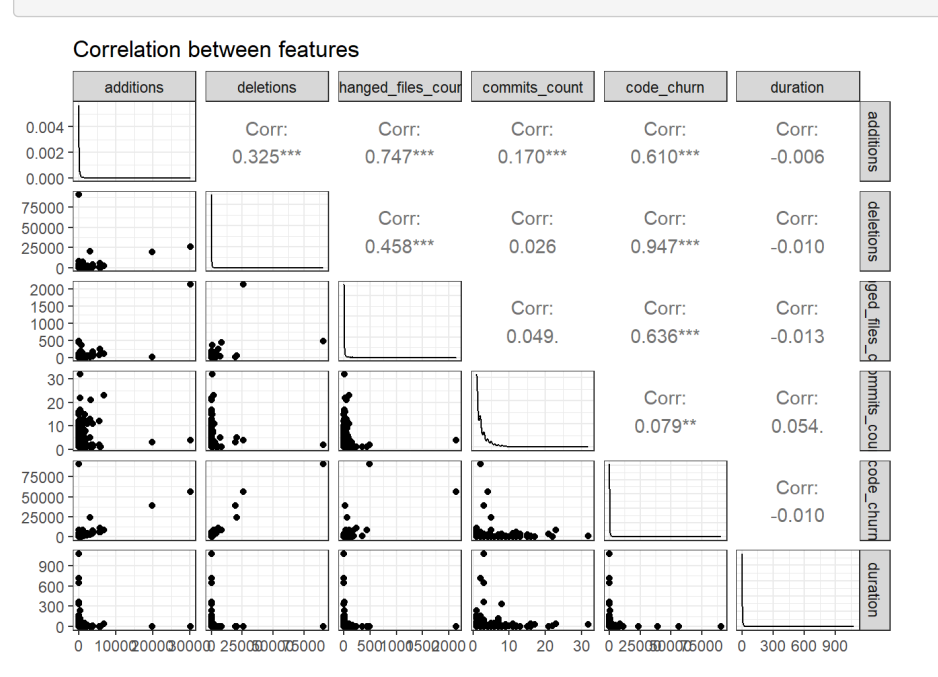
```
prs_characterization %>%
    ggplot(aes(x = duration, y = code_churn, color=repo)) +
    geom_point(alpha=0.75) +
    labs(x = "Duration", y = "Code churn") +
    ggtitle("Code churn per duration")
```



```
prs_characterization %>%
    ggplot(aes(x = changed_files_count, y = code_churn, color=repo)) +
    geom_point(alpha=0.75) +
    labs(x = "Changed files count", y = "Code churn") +
    ggtitle("Code churn per changed files count")
```



Correlations

```
prs_characterization %>%
    select(additions, deletions, changed_files_count, commits_count, code_churn, duration) %>%
    ggpairs(title="Correlation between features")
```